

Principal Component Analysis (PCA)

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What is PCA?

- ▶ Principal Component Analysis (PCA) is a dimensionality reduction technique.
- ▶ It transforms the original variables into a new set of uncorrelated variables (principal components).
- ▶ These components capture the maximum variance in the data.

Why Use PCA?

- ▶ Reduce complexity of high-dimensional data.
- ▶ Remove noise and redundancy.
- ▶ Helps in visualization and improves performance of ML models.

Steps in PCA

1. Standardize the data.
2. Compute the covariance matrix.
3. Calculate eigenvalues and eigenvectors.
4. Sort eigenvectors by decreasing eigenvalues.
5. Select k eigenvectors and project the data.
6. (Optional) Reconstruct original data from the projection.

Example Dataset

Consider the following dataset with 3 features and 6 observations:

Obs	X_1	X_2	X_3
1	2	0	1
2	4	0	3
3	6	0	5
4	10	1	9
5	12	1	11
6	14	1	13

Step 1: Standardize the Data

We subtract the mean and divide by standard deviation.

- ▶ Mean of X_1 : $\mu_1 = 8$ Std Dev: $\sigma_1 = 4.32$
- ▶ Mean of X_2 : $\mu_2 = 0.5$ Std Dev: $\sigma_2 = 0.55$
- ▶ Mean of X_3 : $\mu_3 = 7$ Std Dev: $\sigma_3 = 4.32$

Standardized $Z_i = \frac{X_i - \mu_i}{\sigma_i}$

Step 2: Compute Covariance Matrix

Let Z be the standardized data matrix.

$$C = \frac{1}{n-1} Z^T Z$$

- ▶ Covariance matrix will be 3×3
- ▶ Each C_{ij} is $\text{Cov}(X_i, X_j)$

Step 3: Compute Eigenvalues and Eigenvectors (Part 1)

Given the covariance matrix C :

$$C = \begin{bmatrix} 0.56 & 0.16 & -0.10 \\ 0.16 & 0.56 & 0.02 \\ -0.10 & 0.02 & 0.56 \end{bmatrix}$$

We need to solve the characteristic equation:

$$\det(C - \lambda I) = 0$$

This means subtracting λ from each diagonal element and computing the determinant:

$$\begin{vmatrix} 0.56 - \lambda & 0.16 & -0.10 \\ 0.16 & 0.56 - \lambda & 0.02 \\ -0.10 & 0.02 & 0.56 - \lambda \end{vmatrix} = 0$$

This results in a cubic polynomial in λ . Solving this equation gives the eigenvalues:

$$\lambda_1 = 0.70, \quad \lambda_2 = 0.55, \quad \lambda_3 = 0.43$$

Note: These values are approximations.

Step 3: Compute Eigenvalues and Eigenvectors (Part 2)

For each eigenvalue λ , solve:

$$(C - \lambda I)\vec{v} = 0$$

This yields the eigenvectors:

- ▶ For $\lambda_1 = 0.70$, $\vec{v}_1 = [0.707, 0.707, 0]$ (Principal Component 1)
- ▶ For $\lambda_2 = 0.55$, $\vec{v}_2 = [-0.408, 0.408, 0.816]$
- ▶ For $\lambda_3 = 0.43$, $\vec{v}_3 = [0.577, -0.577, 0.577]$

Interpretation:

- ▶ Eigenvectors represent the directions of maximum variance.
- ▶ Eigenvalues indicate how much variance is along each direction.
- ▶ The eigenvector with the largest eigenvalue is the first principal component.

Step 4: Select Principal Components

- ▶ Rank eigenvectors by descending eigenvalues.
- ▶ Choose top k eigenvectors to form matrix W .
- ▶ W defines new k -dimensional space.

$$Z_{proj} = ZW$$

Step 5: Project Data

Transform original data:

$$\text{Projected Data} = ZW \text{ (dimension: } n \times k \text{)}$$

You now have a reduced representation of the original data.

Step 6: Reconstruct Original Data

- ▶ Reconstruct:

$$\hat{Z} = Z_{proj} W^T$$

- ▶ Undo standardization:

$$\hat{X}_i = \hat{Z}_i \cdot \sigma_i + \mu_i$$

Used to approximate original data from reduced dimensions.

Summary

- ▶ PCA reduces dimensions while retaining variance.
- ▶ It uses linear algebra (covariance, eigenvectors).
- ▶ Data must be standardized before PCA.
- ▶ New axes (principal components) are orthogonal.